

Translated

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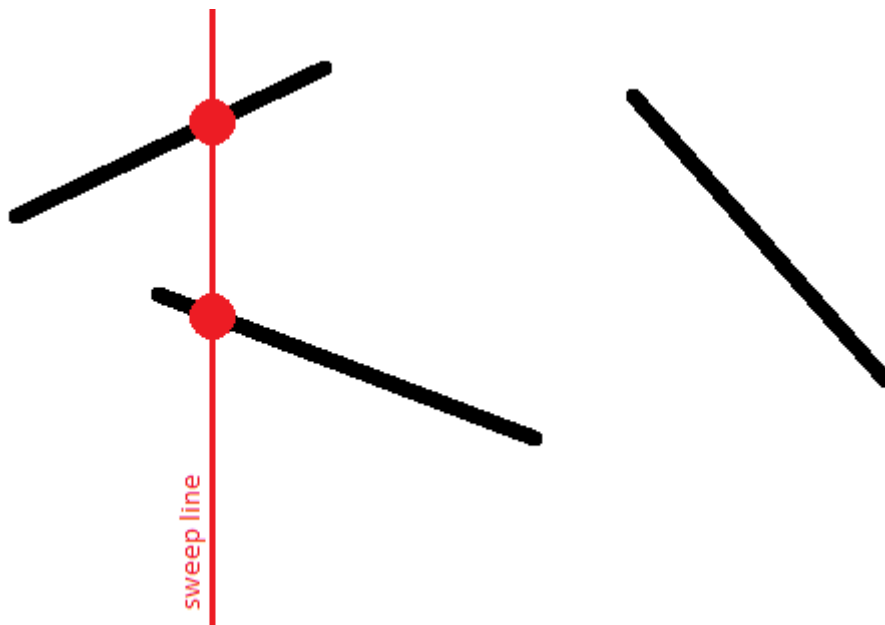
Search for a pair of intersecting segments

Given n line segments on the plane. It is required to check whether at least two of them intersect with each other. If the answer is yes, then print this pair of intersecting segments; it is enough to choose any of them among several answers.

The naive solution algorithm is to iterate over all pairs of segments in $O(n^2)$ and check for each pair whether they intersect or not. This article describes an algorithm with the runtime time $O(n \log n)$, which is based on the **sweep line algorithm**.

Algorithm

Let's draw a vertical line $x = -\infty$ mentally and start moving this line to the right. In the course of its movement, this line will meet with segments, and at each time a segment intersects with our line it intersects in exactly one point (we will assume that there are no vertical segments).



Thus, for each segment, at some point in time, its point will appear on the sweep line, then with the movement of the line, this point will move, and finally, at some point, the segment will disappear from the line.